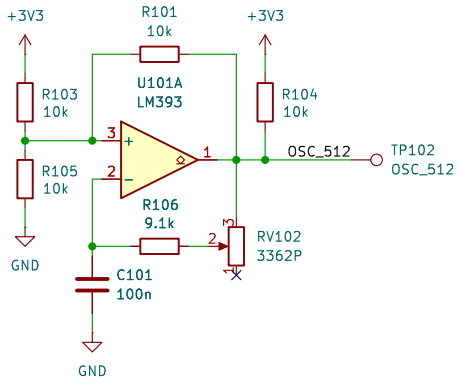
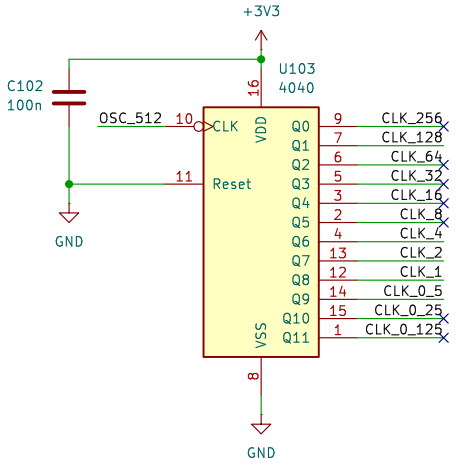
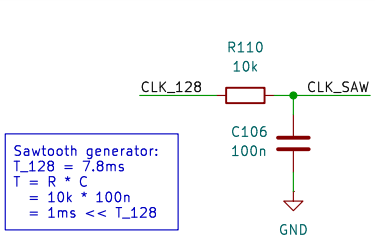
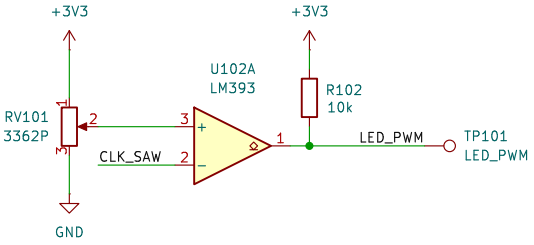


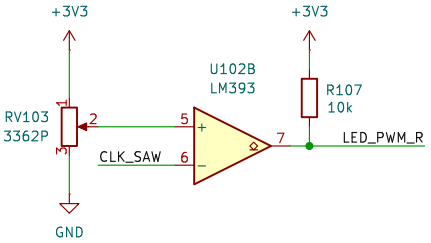
Square signal generator:
F = 512 Hz => T = 1.95 ms
T = 2 * ln(2) * RC
R = 14 kOhms / C = 100n
=> LM393 propagation time to take into account

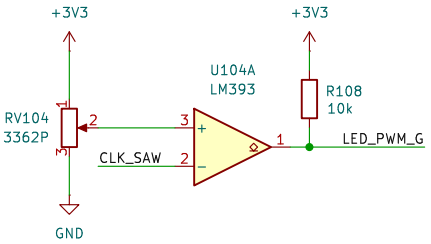


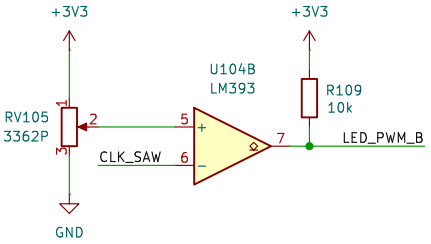


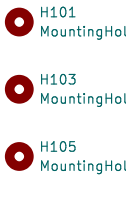
Sawtooth generator:
T_128 = 7.8ms
T = R * C
= 10k * 100n
= 1ms << T_128

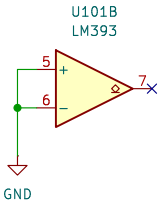


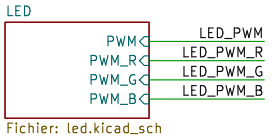


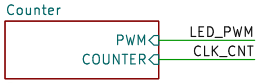




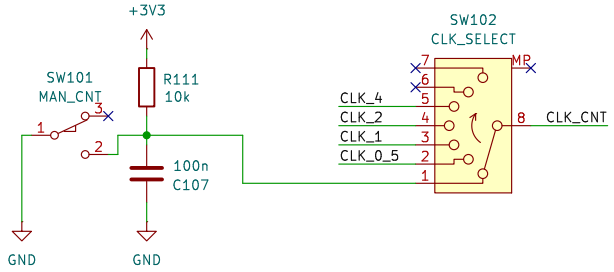


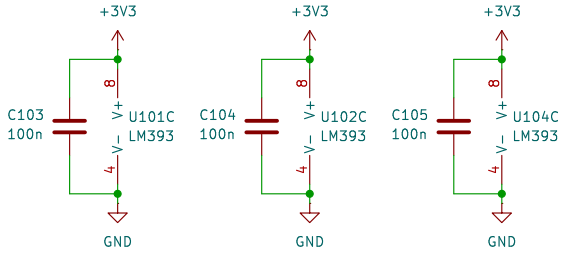








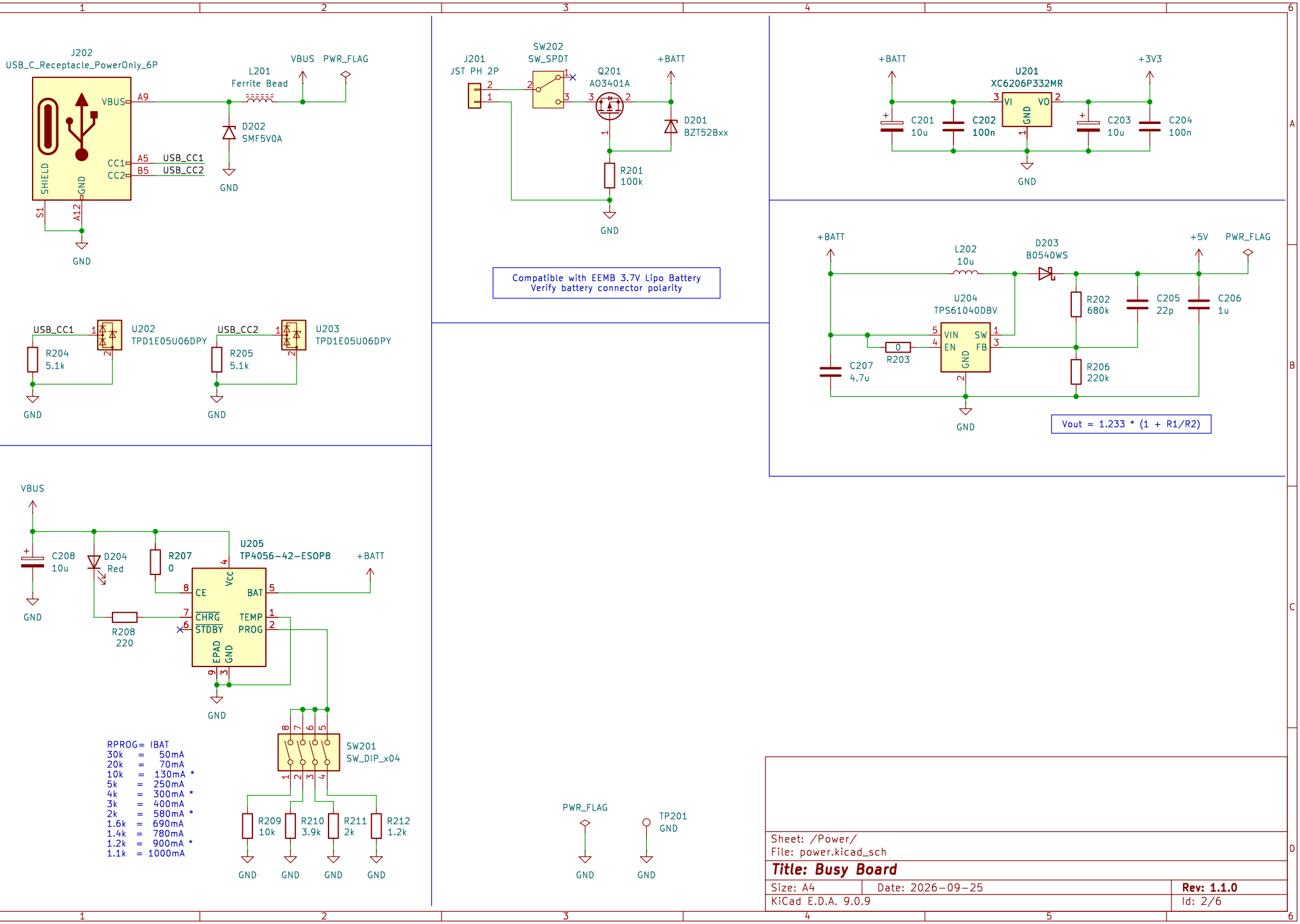


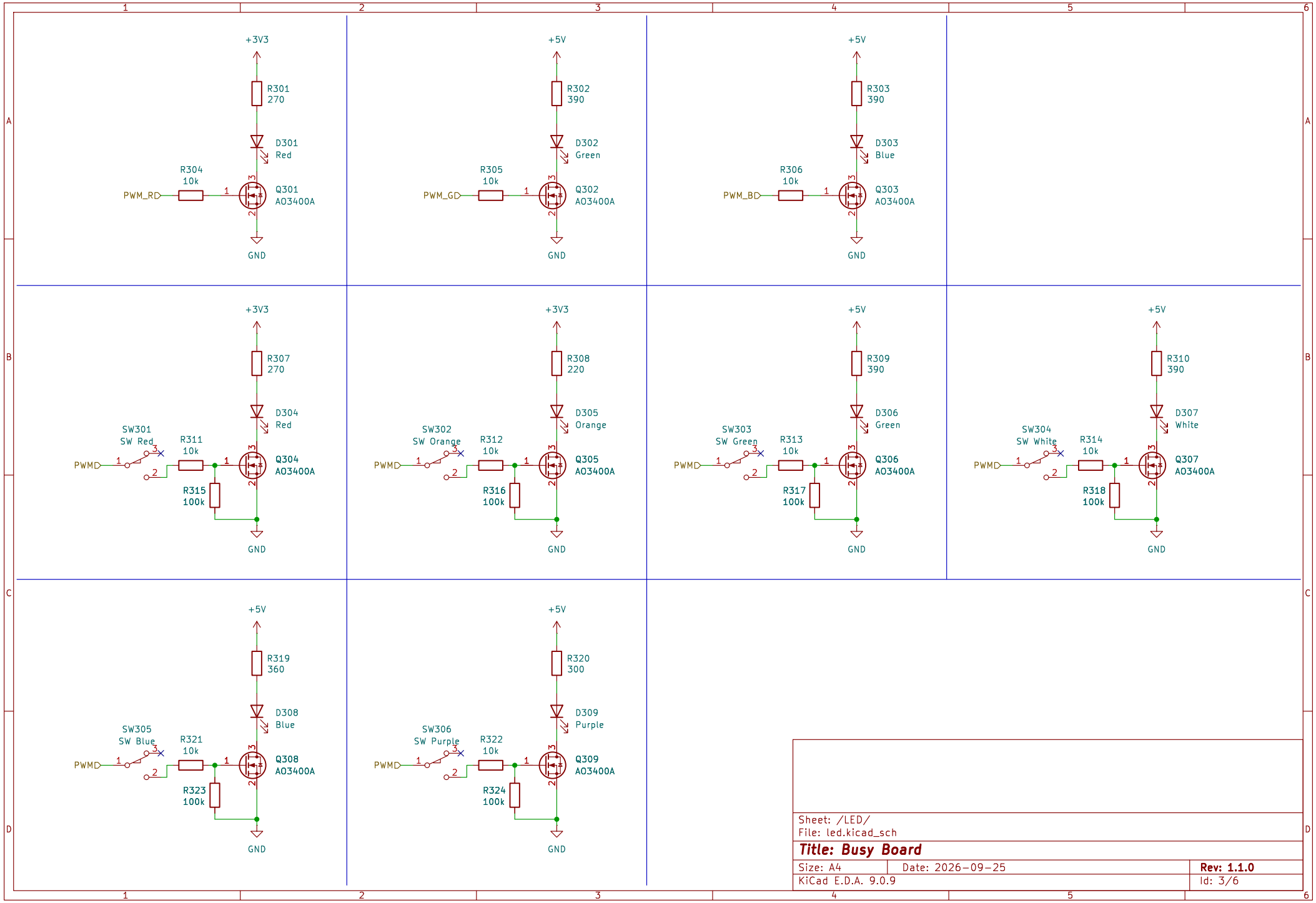


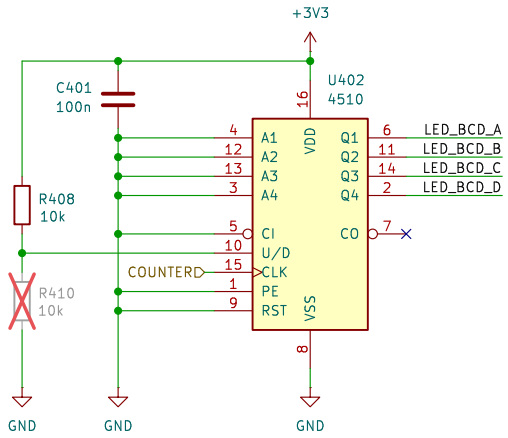
Sheet: /
File: busy-board.kicad_sch

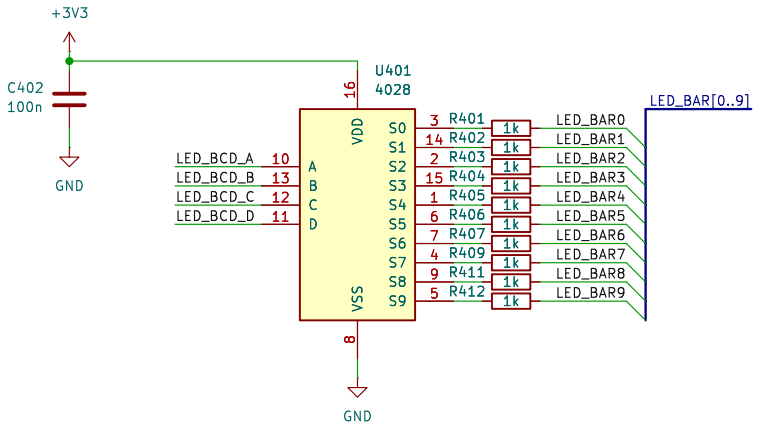
Title: Busy Board

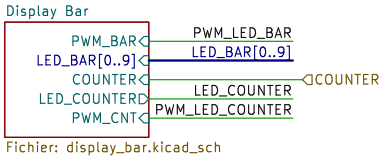
Size: A4	Date: 2026-09-25	**Rev: 1.1.0**
KiCad E.D.A. 9.0.9		Id: 1/6

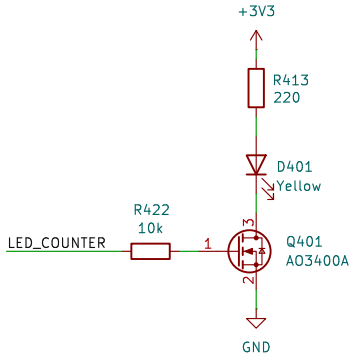

J202
USB_C_Receptacle_PowerOnly_6P
VBUS
A9
L201
Ferrite Bead
VBUS
PWR_FLAG
D202
SMF5V0A
CC1
CC2
A5
USB_CC1
B5
USB_CC2
SHIELD
GND
S1
A12
GND
J201
JST PH 2P
SW202
SW_SPDT
Q201
AO3401A
+BATT
D201
BZT52Bxx
R201
100k
GND
Compatible with EEMB 3.7V Lipo Battery
Verify battery connector polarity
+BATT
U201
XC6206P332MR
+3V3
VI
VO
GND
C201
10u
C202
100n
C203
10u
C204
100n
GND
USB_CC1
U202
TPD1E05U06DPY
R204
5.1k
GND
USB_CC2
U203
TPD1E05U06DPY
R205
5.1k
GND
+BATT
L202
10u
D203
B0540WS
+5V
PWR_FLAG
U204
TPS61040DBV
VIN
SW
EN
FB
GND
R202
680k
C205
22p
C206
1u
0
R203
C207
4.7u
R206
220k
GND
Vout = 1.233 * (1 + R1/R2)
VBUS
C208
10u
D204
Red
R207
0
U205
TP4056-42-ESOP8
+BATT
CE
VCC
BAT
CHRG
STDBY
TEMP
PROG
EPAD
GND
R208
220
GND
RPROG= IBAT
30k = 50mA
20k = 70mA
10k = 130mA *
5k = 250mA
4k = 300mA *
3k = 400mA
2k = 580mA *
1.6k = 690mA
1.4k = 780mA
1.2k = 900mA *
1.1k = 1000mA
SW201
SW_DIP_x04
R209
10k
R210
3.9k
R211
2k
R212
1.2k
GND
GND
GND
GND
PWR_FLAG
GND
TP201
GND
GND
Sheet: /Power/
File: power.kicad_sch
Title: Busy Board
Size: A4
Date: 2026-09-25
Rev: 1.1.0
KiCad E.D.A. 9.0.9
Id: 2/6

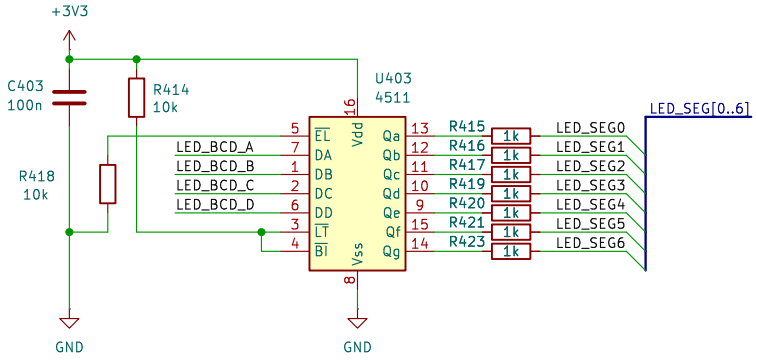

+3V3
R301
270
D301
Red
R304
10k
PWM_R
Q301
AO3400A
GND
+5V
R302
390
D302
Green
R305
10k
PWM_G
Q302
AO3400A
GND
R303
390
D303
Blue
R306
10k
PWM_B
Q303
AO3400A
GND
+3V3
R307
270
D304
Red
SW301
SW Red
R311
10k
PWM
Q304
AO3400A
R315
100k
GND
+3V3
R308
220
D305
Orange
SW302
SW Orange
R312
10k
PWM
Q305
AO3400A
R316
100k
GND
+5V
R309
390
D306
Green
SW303
SW Green
R313
10k
PWM
Q306
AO3400A
R317
100k
GND
+5V
R310
390
D307
White
SW304
SW White
R314
10k
PWM
Q307
AO3400A
R318
100k
GND
+5V
R319
360
D308
Blue
SW305
SW Blue
R321
10k
PWM
Q308
AO3400A
R323
100k
GND
+5V
R320
300
D309
Purple
SW306
SW Purple
R322
10k
PWM
Q309
AO3400A
R324
100k
GND
Sheet: /LED/
File: led.kicad_sch
Title: Busy Board
Size: A4
Date: 2026-09-25
Rev: 1.1.0
KiCad E.D.A. 9.0.9
Id: 3/6

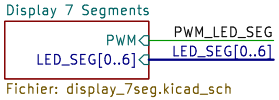


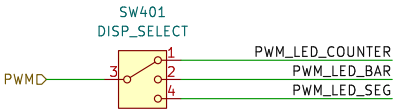








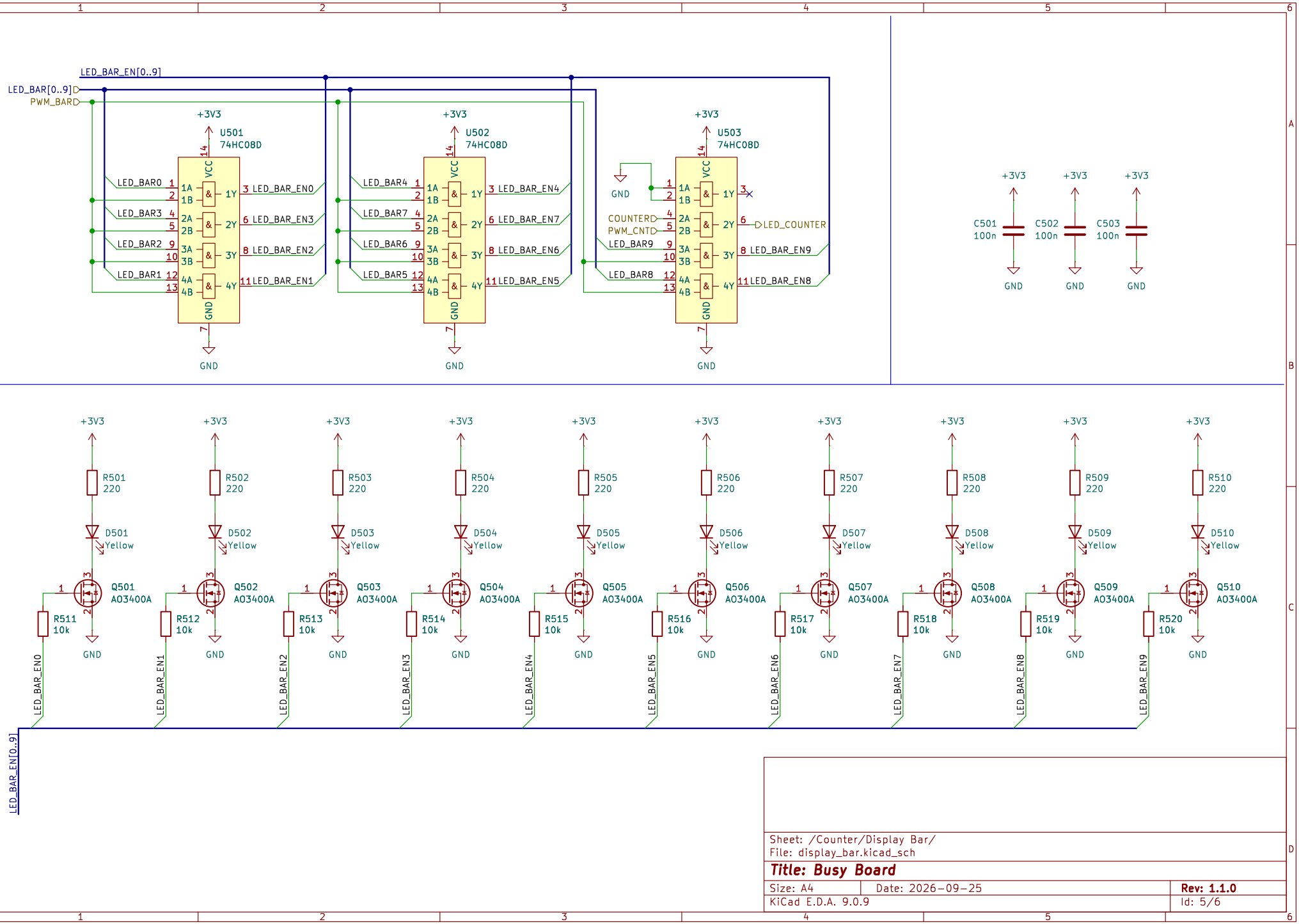




Sheet: /Counter/
File: counter.kicad_sch

Title: Busy Board

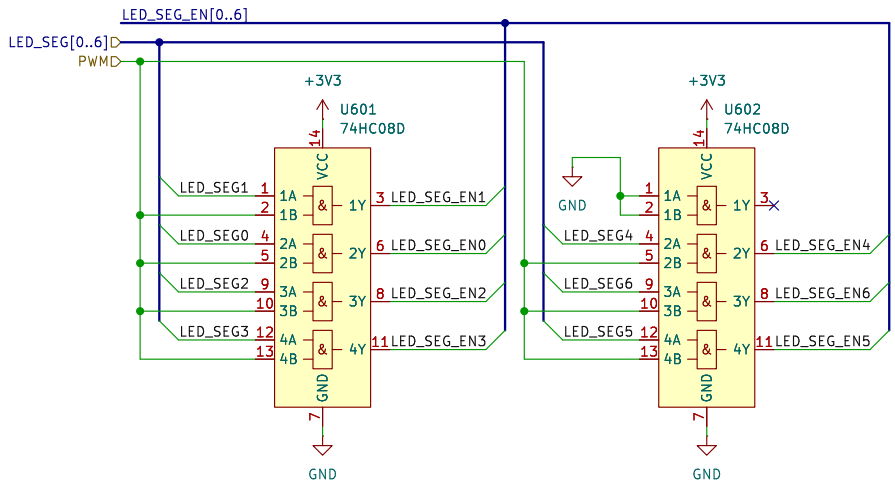
Size: A4	Date: 2026-09-25	**Rev: 1.1.0**
KiCad E.D.A. 9.0.9		Id: 4/6

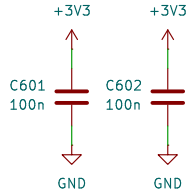


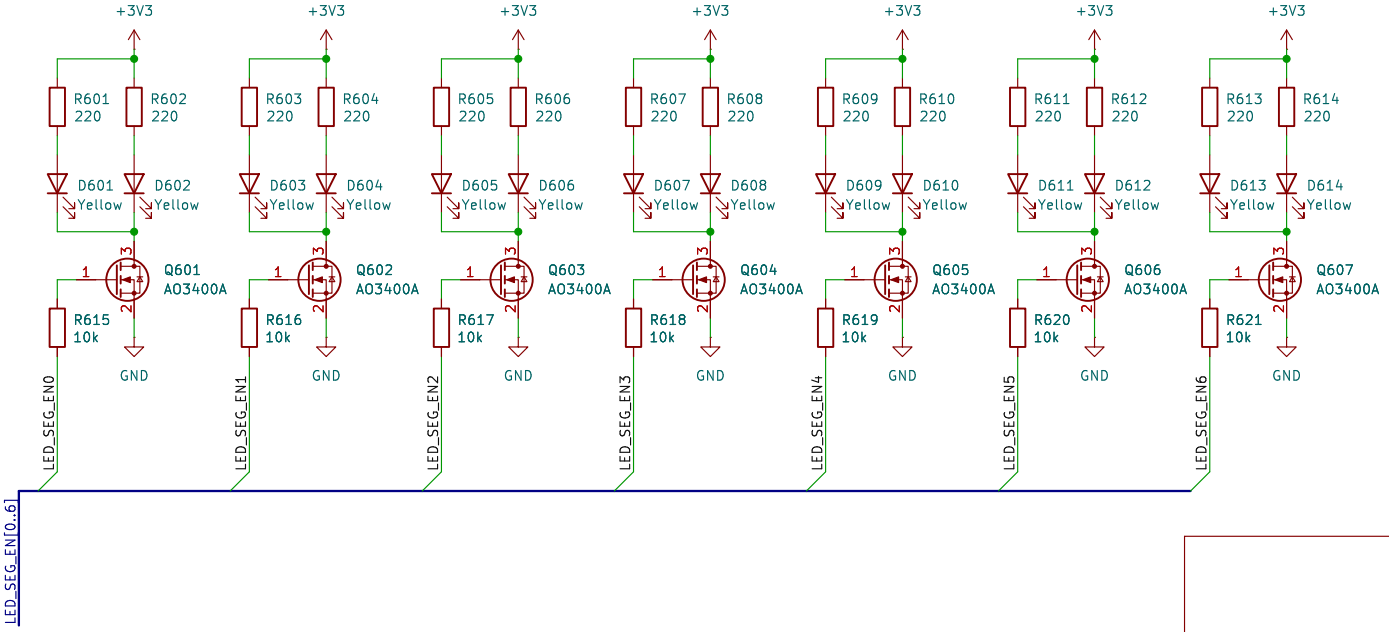
Sheet: /Counter/Display Bar/
File: display_bar.kicad_sch

Title: Busy Board

Size: A4	Date: 2026-09-25	**Rev: 1.1.0**
KiCad E.D.A. 9.0.9		Id: 5/6







Sheet: /Counter/Display 7 Segments/
File: display_7seg.kicad_sch

Title: Busy Board

Size: A4	Date: 2026-09-25	**Rev: 1.1.0**
KiCad E.D.A. 9.0.9		Id: 6/6